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A family of large language models for materials research with insights into model adaptability in continued pretraining

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Materials discovery and development are critical for addressing global challenges in renewable energy, sustainability, and advanced technology. Large language models (LLMs) offer unprecedented opportunities to accelerate materials research, yet their effective deployment requires domain-specific adaptation. Here we present large language models for materials (LLaMat), a family of foundational models for materials science, developed through continued pretraining of LLaMA models on 30 billion tokens derived from approximately 4 million materials science publications and crystallographic data. To develop a materials copilot, the models were adapted by instruction and task fine-tuning on 175,000 materials science question-answering pairs. Through evaluation across 42 tasks covering the entire spectrum of materials research, spanning natural language processing, structured information extraction and crystal generation, we demonstrate that LLaMat consistently outperforms state-of-the-art commercial LLMs (Claude, GPT and Gemini) while maintaining general linguistic capabilities. Beyond demonstrating the effectiveness of domain adaptation for practically deployable materials research copilots, our findings also reveal fundamental insights about LLM adaptation that may influence the development of specialized scientific artificial intelligence systems across domains. For instance, we identify increasing rigidity to domain adaptation in extensively pretrained LLMs such as LLaMA-3. This consistent pattern observed across the experiments suggests a previously unidentified 'adaptation rigidity', where overtrained LLMs exhibit increasing rigidity to domain adaptation.

Materials innovation can potentially address ten of the seventeen United Nations Sustainable Development Goals through advances in sustainable energy systems, advanced electronics and environmentally conscious manufacturing. This imperative for accelerated materials discovery coincides with an unprecedented expansion in the scientific literature—exceeding 114 million journal publications[1]—presenting both opportunities and challenges for materials informatics[2–4]. Materials researchers urgently need artificial intelligence (AI) copilots that can effectively process vast amounts of unstructured text data and semi-structured tables to extract actionable insights from this information deluge.

Large language models (LLMs), also referred to as foundation models, have demonstrated remarkable capabilities in text processing, analysis and generation[5]. In materials science, LLMs have shown promise across the entire research workflow, including (1) rapid literature-based identification of materials[6,7] and synthesis pathways[8],

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(2) in silico crystal structure generation[9–11], (3) autonomous experimental planning[12–14] and (4) results analysis[15,16]. Recent advances[17–20] have demonstrated the efficacy of LLMs in materials concept comprehension, domain-specific query resolution[19,21,22] and simulation code generation[15]. However, critical analyses reveal that general-purpose LLMs often fail at domain-specific challenges, including correctly interpreting scientific phenomena such as physical laws or theories, specialized terminologies[4,15,19,23] and crystal structures[9,24].

Creating effective AI copilots for materials research requires specialized domain adaptation to address these limitations in materials-specific information processing[4]. Although initial task-specific adaptations have yielded promising breakthroughs in structured information extraction[17], materials-specific natural language processing[17,25,26], experimental data analysis[23,27] and crystal structure generation[9–11,28], the field lacks a unified foundation model that integrates these capabilities into a comprehensive assistant for materials researchers.

Here we introduce large language model for materials (LLaMat)—a family of domain-adapted language models designed to serve as comprehensive copilots for materials research. Through a systematic approach combining continued pretraining on 30 billion tokens of materials literature, instruction fine-tuning on 100,000+ hand-curated materials science question-answering pairs and task-specific fine-tuning, LLaMat integrates capabilities for the materials science workflow spanning scientific literature comprehension, information extraction and crystal generation. Our comprehensive evaluation across 42 tasks (Supplementary Section A) covering the entire materials research spectrum demonstrates that LLaMat consistently outperforms commercial LLMs (Claude, GPT and Gemini) while maintaining general linguistic capabilities. Beyond developing an effective materials research copilot, our systematic cross-domain assessment reveals unexpected insights into LLM adaptation dynamics that may influence the development of specialized scientific AI systems across disciplines.

Results

LLaMat: a family of LLMs for materials

To develop LLaMat, we systematically embedded materials domain knowledge on LLaMA base models—specifically LLaMA-2-7B[29] and LLaMA-3-8B[30], hereafter referred to as LLaMA-2 and LLaMA-3, respectively (Supplementary Section C for their architectures). Although larger LLaMA variants such as the 70B models could yield superior performance, our model selection optimizes the balance between computational demands for training and inference, available pretraining data volume and practical deployment considerations for the materials research community. LLaMat is developed through a rigorously designed three-stage process (Fig. 1 and Methods) aimed at creating a comprehensive materials research copilot capable of addressing diverse challenges across the materials discovery pipeline.

The development began with continued pretraining on the base LLaMA models using our meticulously curated R2CID corpus (Methods, Supplementary Table A.1 and Supplementary Section A) containing over 30 billion tokens of materials science knowledge. This extensive corpus encompasses approximately 4 million peer-reviewed publications (94.43%), crystallographic information files (CIF, 2.499%) and Materials Science Community Discourse (MSCD, 0.019%). To prevent catastrophic forgetting of general linguistic capabilities, we strategically incorporated a 3% subset of RedPajama data-the original training corpus of LLaMA models.

We then implemented two specialized fine-tuning pathways to address complementary aspects of materials research. The first pathway produced LLaMat-Chat, a comprehensive materials research copilot designed to assist with literature analysis, information extraction and structured data interpretation. This variant underwent multistage instruction fine-tuning across domains critical for materials scientists: general English comprehension, mathematical reasoning and materials-specific tasks (Methods and Supplementary Section A.1). The model was further optimized on a unified corpus of materials-relevant downstream tasks (Supplementary Table A.2), resulting in demonstrated proficiency across natural language tasks (named entity recognition (NER), relation classification and text classification) and structured information extraction from scientific text and tables (Supplementary Section A.3). Comprehensive descriptions of the materials science tasks evaluated, including task definitions and their practical applications (Supplementary Section B and Supplementary Table B.1), are provided alongside representative examples of model prompts for each task category (Supplementary Section M).

The second pathway developed LLaMat-CIF—a specialist model for crystal structure analysis and generation—a critical capability for accelerating materials discovery. This variant underwent instruction fine-tuning on a hand-curated dataset of CIF comprising five syntactic and four semantic tasks, followed by parameter-efficient fine-tuning (PEFT) specifically optimized for crystal generation (Methods).

To ensure our models excel across the full spectrum of materials science tasks while maintaining general capabilities, we conducted extensive optimization experiments. For pretraining, we systematically explored dataset combinations by prioritizing materials papers strategically interspersed with RedPajama and CIF data (Methods). For instruction fine-tuning, we developed a balanced curriculum spanning general comprehension (OpenOrca[31]), mathematical reasoning (MathQA[32]) and a diverse suite of materials-specific tasks (Supplementary Section E). The materials datasets included MatSciNLP[33] benchmarks and our hand-curated question-answering resources: MatBookQA (2,069 question-answering pairs focused on fundamental materials concepts), MaScQA [15] (1,022 question-answering pairs covering materials properties and characterization) and MatSciInstruct (170,000 question-answering pairs spanning synthesis, characterization and applications)[34] (Supplementary Section A).

Our systematic evaluation revealed several important insights for developing effective scientific copilots. First, dataset distribution and learning rate optimization (Supplementary Section D) proved crucial for maximizing performance across our comprehensive 42-task benchmark. Notably, the optimal hyperparameters differed between LLaMA-2 and LLaMA-3 architectures (Supplementary Sections D and E), suggesting architecture-specific adaptation requirements. For both models, full-corpus pretraining consistently outperformed intermediate checkpoints across all task categories from basic entity recognition to complex information extraction and crystal structure prediction.

During instruction fine-tuning, we observed striking model-specific behavioural patterns: LLaMA-2 showed substantial improvements after OpenOrca fine-tuning, whereas LLaMA-3 demonstrated unexpectedly minimal gains across both materials science and general language tasks (Supplementary Table E.2). The importance of balanced capabilities became evident when models trained without MathQA[32] exhibited severe degradation in mathematical reasoning-failing at elementary calculations despite maintaining strong performance on materials-specific tasks (Supplementary Table E.3). This finding underscores the critical importance of quantitative reasoning for effective materials science copilots.

Perhaps most surprisingly, we found that after general instruction fine-tuning on OpenOrca and MathQA, additional fine-tuning on materials-specific datasets such as Honeybee[34] yielded minimal performance improvements across our comprehensive benchmark (Supplementary Table E.3). This contrasts with previous work showing improvement compared with Honeybee fine-tuning alone[34], suggesting a fundamental distinction between domain knowledge acquisition and instruction-following capabilities in scientific applications. Through rigorous parametric optimization, we identified Pareto-optimal dataset configurations for each model architecture, effectively maximizing

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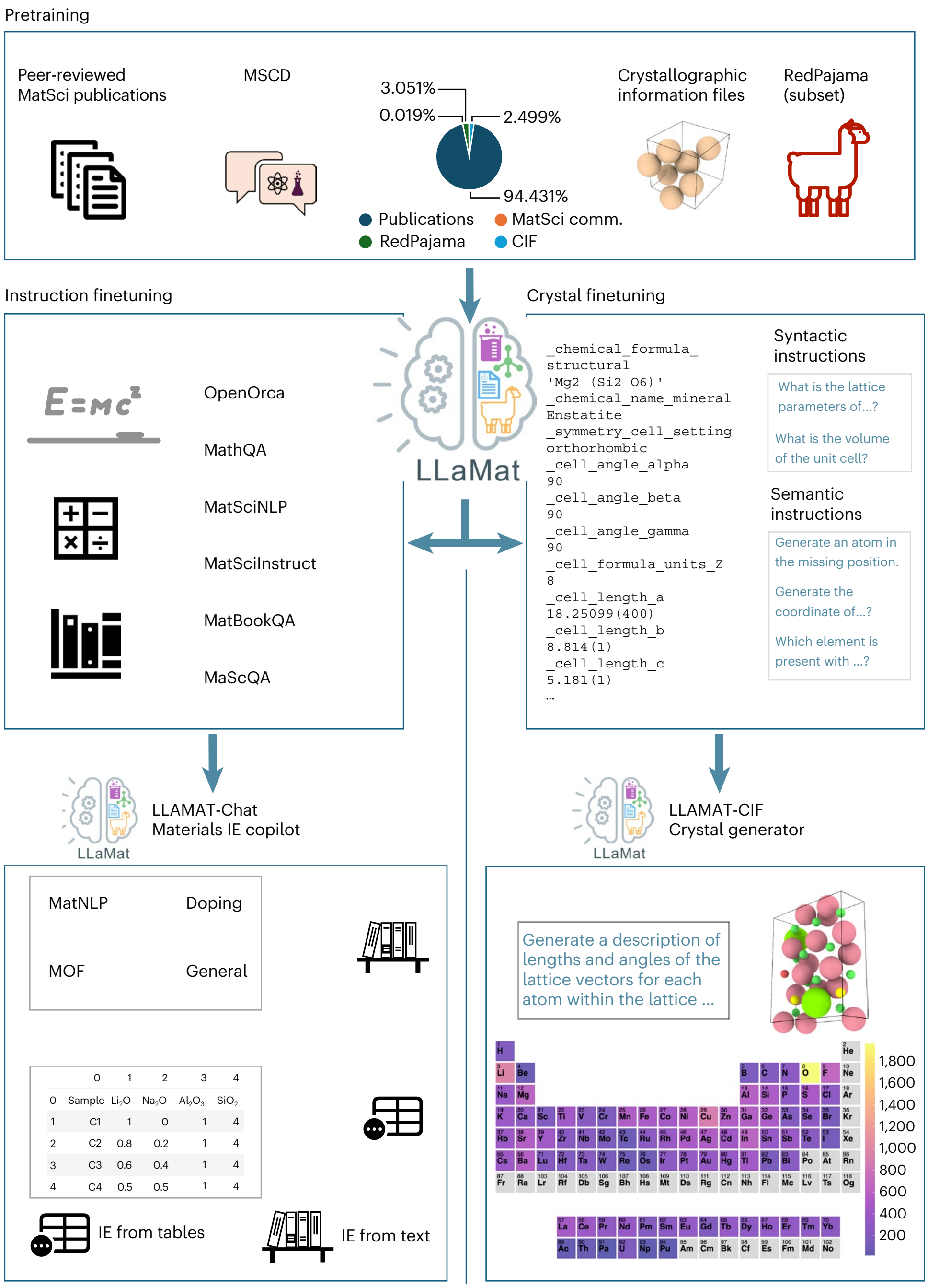

Fig. 1 | Development pipeline and capabilities of LLaMat for comprehensive materials science research. A schematic that illustrates our systematic three-stage development approach for creating a materials research copilot, beginning with continuous pretraining on a specialized 30-billion-token corpus (top), followed by complementary instruction fine-tuning (IFT) pathways (left and right). The pretraining dataset composition is shown in the pie chart, comprising peer-reviewed publications (94.43%), CIF (2.50%), a strategic subset of RedPajama (3.051%), and 0.019% Materials Science Community Discourse (MatSci comm.) to preserve general capabilities. Two distinct fine-tuning pathways yield complementary copilot variants: LLaMat-Chat, a versatile research assistant capable of structured information extraction, materials entity recognition and scientific reasoning across 42 evaluation tasks (left branch) and LLaMat-CIF, specialized in crystal structure analysis and generation with 49.5% thermodynamic stability (right branch). Representative examples demonstrate how these capabilities transform materials science workflows, from literature-based discovery to in silico crystal design.

performance across diverse materials science tasks while maintaining robust general capabilities (Supplementary Section E).

Materials research copilot performance

To validate LLaMat's effectiveness as a comprehensive materials research copilot, we conducted systematic evaluations across two complementary domains essential for materials discovery: materials' natural language processing (MatNLP) and materials structured information extraction (MatSIE). These evaluations assess the model's ability to understand complex scientific concepts and transform unstructured literature into actionable insights-capabilities critical for accelerating the materials discovery pipeline.

Materials language processing. The MatNLP benchmark evaluates LLaMat's ability to interpret and reason about materials science literature through fourteen diverse tasks spanning three fundamental NLP

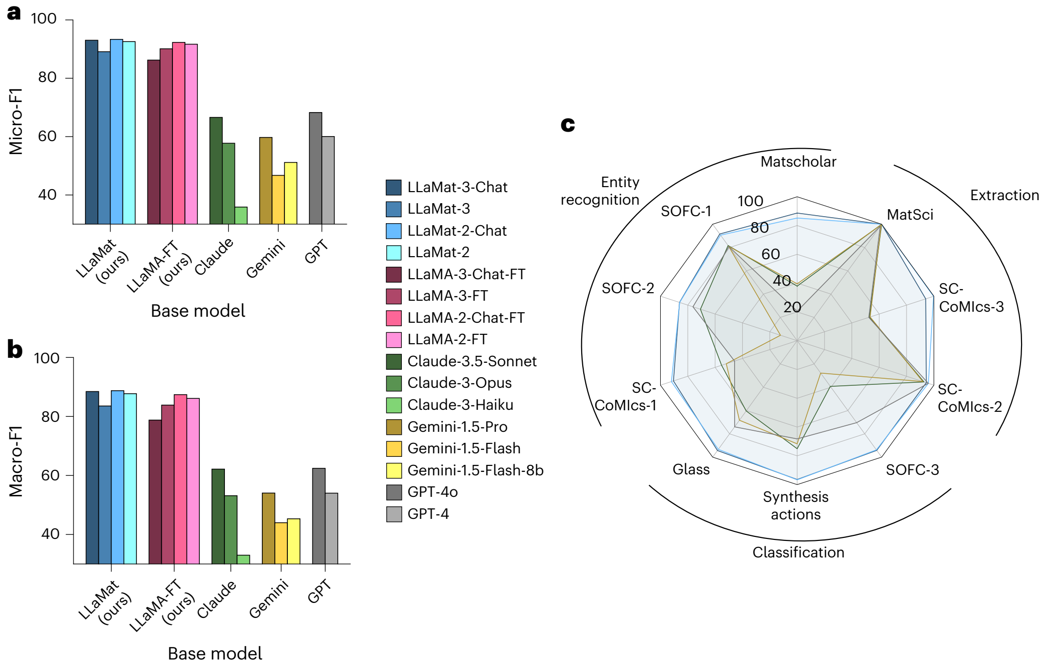

Fig. 2 | Performance evaluation of LLaMat as a materials research copilot compared with commercial LLMs and baseline models. a,**b**, Micro-F1 (**a**) and Macro-F1 (**b**) scores demonstrating LLaMat's superior performance across our MatNLP benchmark comprising 14 MatSci tasks. **c**, Task-specific performance across critical capabilities, including entity recognition, relation extraction and classification tasks essential for automated literature analysis. Only the top-performing models from each family are included: LLaMat-3-chat, LLaMat-2-chat, Claude-3.5-Sonnet, Gemini-1.5-Pro, and GPT-4o. Higher scores indicate better performance in extracting domain-specific information (for example, synthesis protocols and characterization methods), identifying relationships between materials entities and classifying scientific text-capabilities that directly accelerate materials discovery. Results demonstrate that our domain adaptation approach creates more effective materials research copilots than commercial LLMs.

families: entity recognition, extraction and classification. Our evaluation framework comprises ten materials-specific and four general English datasets, totalling 14,579 test instances designed to systematically probe different aspects of scientific language understanding. These tasks assess capabilities essential for materials research automation, including extracting synthesis protocols, identifying characterization methods, recognizing application-specific entities, classifying materials-related documents and comprehending entity relationships. The complementary English datasets ensure the model maintains general language proficiency through question-answering and multiple-choice tasks.

We compared the performance of LLaMat-2 and LLaMat-3 models and their chat variants against their respective base models and leading commercial LLMs (GPT, Claude and Gemini). For rigorous comparison, we also fine-tuned both pretrained and chat variants of LLaMA-2 and LLaMA-3 on identical downstream task datasets. As shown in Fig. 2, LLaMat models significantly outperform both their base models and commercial alternatives. The micro and macro F1 scores (Figs. 2a,b) reveal LLaMat-3-Chat's superior overall performance compared with non-chat variants, validating our domain-specific continued pretraining and instruction fine-tuning strategy. Notably, commercial LLMs demonstrate significantly inferior performance on materials-specific tasks (Supplementary Section F), despite their substantially larger parameter counts and more extensive pretraining. All evaluations were conducted with temperature set to 0 to ensure deterministic outputs.

The radar plot (Fig. 2c) provides a granular analysis of micro-F1 scores across MatNLP dataset subsets, comparing top-performing models from each family. Most notably, LLaMat-3-Chat demonstrates consistent performance advantages across diverse MatSci tasks, including entity recognition, classification and extraction tasks, with LLaMat-2-Chat ranked second. All state-of-the-art commercial models-including Claude-3.5-Sonnet, Gemini-1.5-Pro and GPT-4o-exhibit significantly inferior performance, establishing the efficacy of our domain-adapted models for practical materials science applications. This task-specific analysis confirms that LLaMat models excel precisely in the capabilities most critical for automating materials research workflows.

Our comprehensive analysis also revealed a surprising pattern in domain adaptation capabilities. Although LLaMat-3 variants exhibit greater relative improvement from their base models compared with LLaMat-2, the fine-tuned LLaMAat2 models consistently outperform their LLaMA-3 counterparts across materials-specific tasks. This counterintuitive pattern persists even in continued pretraining models without instruction fine-tuning, where LLaMat-2 demonstrates superior materials science performance despite starting from a weaker base model. This observation suggests a previously unidentified domain adaptation limitation in extensively pretrained models such as LLaMA-3, possibly stemming from their significantly larger pretraining corpus (approximately three orders of magnitude more data) despite superior base capabilities. We term this phenomenon 'adaptation rigidity', a finding that consistently recurs across our evaluation tasks and highlights the complex interplay between model architecture, pretraining scale and domain adaptation effectiveness[35,36].

Structured information extraction from text. Materials literature contains vast knowledge about compositions, synthesis protocols and property relationships embedded within unstructured text. Converting this information into structured formats represents a critical step for accelerating materials discovery workflows. Traditional approaches require extensive manual annotation and specialized models for each extraction task-a particularly challenging problem in domains like doping studies and metal-organic frameworks (MOFs), where precise extraction of chemical compositions, structural relationships and functional properties is essential. Although recent studies have

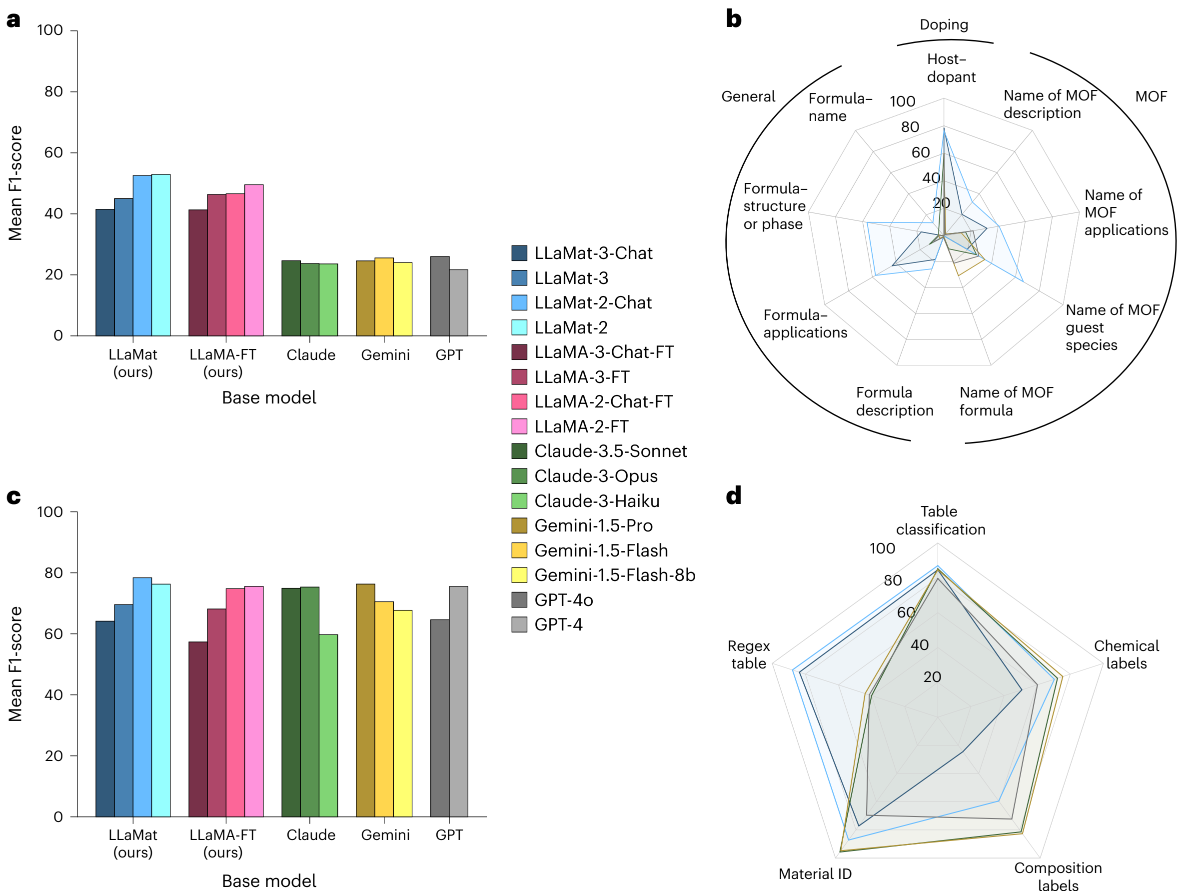

Fig. 3 | Performance evaluation of structured information extraction capabilities across MatSci subdomains. **a**, Mean F1 score across all our structured information extraction tasks in doping, MOFs and general material science. **b**, F1 score across all relation extraction tasks. **c**, Mean accuracy across all material science table data extraction tasks. **d**, F1 score for individual tasks in table data extraction. Only the top models from each family are included in the radar plots, LLaMat-3-chat, LLaMat-2-chat, Claude-3.5-Sonnet, Gemini-1.5-Pro and GPT-4o.

demonstrated commercial LLMs' potential for these tasks[17,37], their proprietary nature and associated costs limit scalable deployment across millions of materials science articles, highlighting the need for open-source alternatives optimized for materials research[7].

Building on LLaMatChat's superior performance in materials language processing, we evaluated its structured information extraction capabilities across specialized scientific domains. Figure 3a demonstrates LLaMat-Chat's performance across nine distinct extraction tasks spanning doping studies, MOFs and general materials domains. Our models consistently outperform all commercial LLMs (Claude, GPT and Gemini) despite their significantly larger parameter counts. Complete performance comparisons across all model variants are provided in Supplementary Section F. Both LLaMat-2 and LLaMat-3 chat variants significantly outperform their fine-tuned LLaMA counterparts in critical materials science information extraction tasks, including host–dopant relationships, formula–structure mappings and application-specific information. Figure 3b shows the performance of best-in-class models from each family across all nine MatSIE tasks, with LLaMat-Chat models demonstrating superior capabilities in most categories.

Notably, LLaMat-2-Chat exhibits particularly strong performance in formula–application relationships and host–dopant associations tasks requiring deep materials domain knowledge. This performance pattern further reinforces our observation of the adaptation rigidity phenomenon, where LLaMat-2-Chat demonstrates significantly enhanced capabilities compared with its successor after domain-specific adaptation. This consistent trend across diverse evaluation metrics strengthens our hypothesis about the inverse relationship between the initial pretraining scale and domain adaptation efficacy.

Information extraction from tables. Tables in materials science publications serve as structured repositories of composition-property relationships, yet present unique challenges due to their heterogeneous formats and complex organizational schemas[23,38]. Extracting structured information from these diverse tabular representations requires models with both deep materials domain knowledge and sophisticated format understanding. We evaluated LLaMat's ability to extract meaningful information from materials science tables using a challenging benchmark of 737 tables from peer-reviewed publications[38]. Our assessment focused on five critical capabilities essential for materials research automation: compositional table classification, chemical constituent localization, composition extraction, material identifier recognition and regex-amenable information identification.

As shown in Fig. 3c, LLaMat-2-Chat models achieve superior overall performance, slightly outperforming commercial LLMs. Interestingly, although the performance gap between LLaMat and commercial models is less pronounced for tables than for other tasks, we observe the same recurring adaptation pattern: LLaMat-2 and LLaMA-2 models consistently outperform their third-generation counterparts across all evaluation metrics. This advantage is particularly evident in chemical label identification and composition extraction tasks, further supporting our adaptation rigidity hypothesis and suggesting this phenomenon extends to structured data interpretation. Detailed

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Table 1 | Mean performance improvement of fine-tuned LLaMA models over their base counterparts on MatNLP tasks

Metric	Category	LLaMA-2			LLaMA-3		
		Base	Fine-tuned	%Δ	Base	Fine-tuned	%Δ
Macro-F1	MatNLP	12.36	87.398	+607.10%	32.287	78.775	+143.98%
Micro-F1	MatNLP	14.827	92.295	+522.48%	36.489	86.225	+136.30%

The base models are LLaMA-2-Chat and LLaMA-3-Chat, and the fine-tuned models are LLaMA-2-Chat-FT and LLaMA-3-Chat-FT.

performance metrics and task-specific analyses are provided in Supplementary Section G. Figure 3d presents a task-specific performance comparison of best-in-class models. The analysis reveals an interesting pattern: LLaMat's overall superior performance is primarily driven by excellence in regex tables, where complex domain-specific notations represent materials in tabular format. For other table types, commercial LLMs demonstrate comparable or occasionally superior performance, probably benefiting from extensive exposure to tabular formats in their larger pretraining datasets. This suggests that although domain-specific adaptation significantly enhances LLaMat's performance on materials-specific notation and terminology, further improvements in general table structure understanding could further enhance its capabilities as a materials research copilot.

Adaptation rigidity. To rigorously validate the adaptation rigidity hypothesis observed across our evaluations, we conducted comprehensive comparisons between base and fine-tuned models on all MatNLP and MatSIE tasks (see Supplementary Section L for detailed results). The results reveal a striking domain-dependent pattern: although LLaMA-3-Chat demonstrates superior baseline performance, LLaMA-2 exhibits dramatically larger improvements from domain-specific fine-tuning on materials science tasks. Across MatNLP materials tasks, LLaMA-2-Chat-FT achieves 607.10% improvement in Macro-F1 over its base model, compared with only 143.98% for LLaMA-3-Chat-FT (Table 1). This domain adaptation advantage is particularly pronounced in specialized materials science tasks. On materials-specific tasks such as matscholar NER, LLaMA-2-Chat-FT shows 816.43% improvement versus 584.32% for LLaMA-3-Chat-FT and achieves superior absolute performance on sc_comics NER (90.82% versus 79.58%) and glass classification (92.0% versus 86.67%) (Supplementary Tables L.1 and L.2). Similarly, on MatSIE tasks, LLaMA-2-Chat-FT demonstrates substantially larger gains: 378.0% improvement on formula extraction versus 134.1% for LLaMA-3-Chat-FT and 1464.1% on formula–application relation extraction versus 1047.2% (Supplementary Table L.3).

Intriguingly, this pattern reverses for general English language tasks within the same evaluation framework. On tasks such as SQuAD and HellaSwag, LLaMA-3-Chat-FT achieves superior absolute performance (85.11% and 84.22%) compared with LLaMA-2-Chat-FT (85.02% and 82.28%), despite both models undergoing identical fine-tuning protocols (Supplementary Tables L.1 and L.2). This domain-dependent adaptation behaviour suggests that extensively pretrained models such as LLaMA-3 develop strong priors optimized for general language understanding that paradoxically hinder specialization to domain-specific knowledge, whereas simultaneously maintaining advantages on general tasks. This observation aligns with recent findings on catastrophic overtraining[39], providing empirical evidence that pretraining scale and domain adaptation efficacy exhibit a complex, task-dependent relationship rather than the monotonic improvement typically assumed in model scaling paradigms.

Interpretability analyses using attribution methods (Supplementary Section P) provide mechanistic insights into this phenomenon, revealing that LLaMat-2 develops more focused domain-specific token associations, whereas LLaMat-3 exhibits more distributed, uniform patterns—consistent with our hypothesis that extensive pretraining creates internal representations resistant to targeted specialization.

Interactive and agentic capabilities. Beyond quantitative benchmarks, LLaMat demonstrates practical utility through interactive capabilities for real-world materials research. To this end, we developed an interactive dashboard[40] (Supplementary Section M and Supplementary Fig. M.1) enabling researchers to explore model outputs and compare performance across datasets. This tool illustrates LLaMat's domain understanding, for instance, correctly distinguishing paragraphs about glass surface coatings from bulk glass properties, a classification requiring materials knowledge that challenges domain-general models.

LLaMat-Chat variants exhibit robust conversational abilities for domain-specific enquiries, providing technically accurate responses about materials concepts while supporting follow-up questions essential for scientific enquiry. The model's architecture is also suited for integration into agentic research pipelines, where it can serve as a knowledgeable agent providing materials expertise to broader workflows—interpreting experimental results, suggesting follow-up experiments or providing contextual information during automated synthesis planning (Supplementary Section O and Supplementary Fig. O.1). This transforms LLaMat from a standalone tool into a collaborative research partner for autonomous materials discovery. We provide visualization tools[40] and chat interface and agentic pipeline integration[41] examples in our GitHub repository[42] to facilitate community adoption.

Crystal generation. Recent works have demonstrated the potential of LLMs to generate crystal structures[9–11]. To evaluate crystal generation capabilities of LLaMat, we developed LLaMat-CIF through crystallographic instruction fine-tuning and PEFT for structure generation. The instruction fine-tuning employed a dual-framework approach balancing syntactic understanding of CIF file formats with semantic comprehension of crystal stability principles (see Supplementary Section I for the prompts). Syntactic tasks focused on structural interpretation (atomic frequency quantification, spatial coordinate analysis and crystallographic parameter determination), whereas semantic tasks targeted stability principles (elemental co-occurrence patterns, atomic spatial distributions and stability-determining properties). This methodology generated approximately 7 million instruction–output pairs (6,941,865 training instances and 27,183 validation instances), enabling LLaMat to develop an integrated understanding of both technical file formats and fundamental materials science principles necessary of crystal structures. We emphasize that this experiment serves as a proof of concept demonstrating LLMs' capability to generate syntactically valid CIF files, rather than establishing a state-of-the-art crystal generation model.

Unconditional generation. We evaluated unconditional crystal generation using metrics adapted from the LeMat-GenBench framework[43], which provides standardized benchmarks for assessing structural validity, thermodynamic stability and generative diversity (see 'Crystal generation' section in the Methods). Unconditional generation produces complete CIF structures from scratch using the prompt: 'Below is a description of a bulk material. Generate a description of the lengths and angles of the lattice vectors and then the element type and coordinates for each atom within the lattice'. This unguided approach tests the model's ability to generate chemically and structurally valid CIF autonomously without specifying target elements or structural prototypes.

Table 2 | Unconditional generation

Model	Valid ↑	Unique ↑	Novel ↑	Energy based ↓			Stable ↑	SUN ↑	MSUN ↑
				E_f	E_{hull}	Relaxation r.m.s. deviation			
LLaMat-2-CIF	**76.77**	89.93	58.29	−1.308 ± 3.02	2.37 ± 5.25	0.41 ± 0.57	**214**	**128**	**213**
LLaMat-3-CIF	13.18	**95.22**	**73.5**	1.5 ± 7.25	3.41 ± 5.27	0.85 ± 0.99	16	11	15

The valuation metrics for LLaMat-2 and LLaMat-3 generative crystal models evaluated on 10,000 generated structures each are shown. The direction indicates goodness of results. The bold indicates best performance associated with each metric.

Table 2 presents evaluation results on 10,000 generated structures per model. LLaMat-2-CIF demonstrates the following performance across key benchmarks: 76.77% validity (structures satisfying hard constraints including CIF readability, minimum interatomic distances >0.5 Å and density bounds 0.01–25 g cm^{-3}), 89.93% uniqueness based on Short-BAWL fingerprinting and 58.29% novelty compared with the LeMat-Bulk reference database. Energy-based metrics computed using an ensemble of three machine-learned interatomic potentials (MACE-MP[44], UMA[45] and Orb-v3[46]) show formation energy of −1.308 ± 3.02 eV per atom, energy above hull of 2.37 ± 5.25 eV per atom and relaxation root mean standard deviation (r.m.s. deviation) of 0.41 ± 0.57 Å. Critically, LLaMat-2-CIF achieves 214 structures with the energy above convex hull (E_{hull}) ≤ 0 eV per atom (2.14% of valid structures) and stable, unique and novel (SUN) rate of 128, outperforming LLaMat-3-CIF (16 stable structures, SUN rate of 11).

Notably, LLaMat-2-CIF required only 13,000 generation attempts to produce 10,000 valid structures, whereas LLaMat-3-CIF required approximately 33,000 attempts—a 2.5× efficiency difference reflecting superior structural validity at generation time. This pattern reinforces our adaptation rigidity hypothesis: despite LLaMat-3's stronger baseline capabilities, LLaMat-2 demonstrates superior domain-specific generation after fine-tuning. Additional evaluation using the metrics following ref. 10 is provided in Supplementary Section J, showing LLaMat-2-CIF achieves 49.49% thermodynamic stability compared with 38.0% for the best baseline LLaMA-2 13B model. These metrics enable a direct comparison with prior crystal generation approaches. Detailed structural analysis (Supplementary Figs. J.1 and J.2) reveals that LLaMat-2-CIF generates structures predominantly near ground state energies with compositional landscapes favouring two to four unique elements-distributions aligned with synthetically accessible materials. The complete generated structures (both as-generated and relaxed) are available via Zenodo[47] to enable community validation and analysis.

Conditional generation. To assess conditional generation capabilities, we evaluated LLaMat-CIF models on composition- and space-group-conditioned generation using 9,046 unique test structures from ref. 10. Given target composition and space group specifications, the models generate corresponding crystal structures. Table 3 presents composition and space group matching performance across structural categories.

LLaMat-2-CIF achieves 79.1% overall composition matching with 90.8% extraction success, outperforming LLaMat-3-CIF (5.5% matching, 69.2% extraction)—a 73.6 percentage point gap. For ternary compounds (58.9% of test set), LLaMat-2-CIF achieves 80.7% matching versus 4.7% for LLaMat-3-CIF. Space group matching shows LLaMat-2-CIF achieving 90.8% accuracy for cubic systems and maintaining >70% across all crystal systems. Additional metrics using ref. 10 methodology (Supplementary Table J.1) show LLaMat-2-CIF-C achieves 0.65 composition validity and 0.95 recall with Wasserstein distance N_{el} = 0.129, compared with LLaMat-3-CIF-C's 0.616 validity, 0.93 recall and N_{el} = 0.279.

The performance difference despite identical PEFT protocols reinforces adaptation rigidity across both unconditional and conditional generation tasks. These results demonstrate that LLaMat-CIF models, particularly LLaMA-2-CIF, can generate syntactically valid CIF files with reasonable composition and space group matching capabilities. Complete evaluation methodology, structural analysis (Supplementary Figs. J.1 and J.2) and code are available in Supplementary Section J and our GitHub repository[42].

Discussion

Our results demonstrate that domain-adapted foundational language models can achieve competitive or superior performance compared with significantly larger commercial LLMs on materials-specific tasks across the research workflow. Our comprehensive evaluation across tasks spanning materials-specific natural language processing, structured information extraction from text and tables, and crystal structure generation shows that LLaMaT models consistently outperform general-purpose LLMs such as GPT, Claude and Gemini despite their substantially larger parameter counts. Strategic domain adaptation through continued pretraining and targeted instruction fine-tuning transforms LLMs into specialized scientific copilots without compromising their foundational capabilities. Although commercial models maintain advantages on certain general language tasks and some table interpretation challenges, the substantial performance gains on materials-specific natural language processing and information extraction demonstrate the value of targeted domain adaptation for scientific applications. The successful adaptation of smaller LLaMA family models suggests that domain-specific adaptation of moderate-sized models may provide a more economical and practical solution for scientific applications than relying on general-purpose LLMs.

The LLaMat-CIF models represent a notable advance in crystal structure prediction. A particularly significant finding includes the conditional generation capability, which enables inverse design, that is, generating crystal structures tailored to specified compositions or space groups, as well as extendable to properties. Our results demonstrate that domain-adapted LLMs can serve as powerful tools for the generative modelling of crystals—a capability traditionally requiring specialized physics-based approaches or complex graph neural networks. The models' ability to implicitly learn realistic chemical constraints—evidenced by systematic trends in elemental compositions, reasonable thermodynamic stability and crystal system preferences—opens new possibilities for accelerating materials discovery while maintaining physical and chemical validity. The textual nature of these models also offers unique advantages, potentially enabling researchers to explore synthesis pathways for generated crystal structures through the same interface used for literature analysis, though integration with experimental validation workflows remains an important direction for future work.

A striking finding emerges in the differential performance between model generations. Despite LLaMA-3's superior baseline capabilities, LLaMat-2 variants demonstrate unexpectedly enhanced domain-specific adaptability across multiple tasks, particularly in materials-focused information extraction and crystal structure generation. This raises fundamental questions about the relationship between pretraining scale and domain adaptability through continued pretraining[35,36,39]. This adaptation rigidity phenomenon, reported here for the first time to the best of our knowledge, challenges conventional

Table 3 | Conditional crystal generation performance

Category	Subcategory	Match rate (%)		Extraction success (%)		Performance gap
		LLaMat-2-CIF (L2)	LLaMat-3-CIF (L3)	LLaMat-2-CIF	LLaMat-3-CIF	(L2–L3)
Element count	1 (0.8%)	73.2	29.6	73.2	57.7	43.6
	2 (19.8%)	75.0	5.8	88.4	62.0	69.2
	3 (58.9%)	80.7	4.7	91.9	68.1	76.0
	4 (18.2%)	79.2	6.5	90.8	80.4	72.7
	5 (2.2%)	74.5	7.5	89.0	76.5	67.0
	6+ (0.1%)	75.0	25.	87.5	62.5	50.0
Space group	Cubic (23.8%)	90.8	6.4	93.3	56.6	84.4
	Orthorhombic (19.1%)	74.0	4.5	88.6	77.7	69.5
	Tetragonal (17.3%)	77.1	4.7	90.0	65.3	72.4
	Monoclinic (14.2%)	71.7	5.4	90.6	83.2	66.3
	Hexagonal (10.8%)	79.9	4.7	90.6	60.7	75.2
	Trigonal (10.7%)	76.8	6.8	91.1	71.8	70.0
	Triclinic (4.2%)	72.3	8.5	90.4	87.2	63.8
Overall	All categories	79.1	5.5	90.8	69.2	73.6

A comparison of LLaMat-2-CIF and LLaMat-3-CIF on composition and space group matching across 9,046 test structures is shown. The match rate indicates percentage of generated structures matching target specifications, and extraction success indicates percentage of parseable outputs. The performance gap shows the L2–L3 difference in match rate. The percentages in subcategories indicate proportion of test set.

scaling assumptions in language model development (Supplementary Section L). We hypothesize that the loss landscape[48] in extensively pretrained LLaMA-3 models may have fundamentally different characteristics compared with LLaMat-2, potentially creating barriers to effective domain adaptation. Our interpretability analyses (Supplementary Section P) provide further insights into this mechanism. This observation suggests the need to re-evaluate scaling strategies for domain-specific AI applications and motivates future work examining whether similar patterns emerge across other model families and domains.

Looking ahead, this work establishes a foundation for integrating AI systems into materials research workflows. The complementary capabilities demonstrated across our evaluation—from literature analysis and structured information extraction to crystal structure generation—form a powerful integrated toolkit that addresses multiple bottlenecks in the materials discovery pipeline[4]. These capabilities are particularly well-suited for incorporation into agentic workflows, where LLaMat models could autonomously plan experiments, search the literature for relevant precedents, extract structured information to inform decisions and generate candidate structures for further investigation—all while maintaining a coherent research strategy. Future development should focus on enhancing model robustness, validating performance through human expert evaluation, expanding capabilities to broader materials science applications including experimental design and multimodal reasoning and developing theoretical frameworks for understanding domain adaptation in LLMs. Critically, investigating whether the adaptation patterns we observe generalize across model families and scientific domains will determine whether our findings reflect fundamental principles or architecture-specific phenomena. The insights gained from this study may inform fundamental principles for developing specialized AI systems across scientific domains, potentially transforming how researchers leverage language models for scientific discovery.

Methods

Dataset preparation

Pretraining dataset R2CID. The performance of foundation models is fundamentally determined by their pretraining dataset composition, necessitating meticulous curation of the constituent data sources. Our pretraining dataset, designated R2CID, integrates three distinct components: scientific literature from materials research publications, a curated subset of RedPajama (the original pretraining corpus for LLaMA models) and CIF. The scientific literature provides comprehensive materials characterization and synthesis protocols, whereas the RedPajama subset helps prevent the catastrophic forgetting of the English language processing capabilities. The CIF datasets provide information on crystal structures, including atomic positions, lattice parameters and symmetry operations. This tripartite combination enabled continued pretraining to generate the LLaMat models. The specific composition and characteristics of each dataset component are detailed below.

(1) Research papers. Our corpus comprises over 4 million peer-reviewed articles sourced from approximately 500 Elsevier[49] and 300 Springer[50] journals. Selection criteria included full-text accessibility in XML format for Elsevier publications and HTML format for Springer publications. Journal selection was made manually based on the relevance to the materials domain. Article acquisition utilized the CrossRef API[1] to extract digital object identifiers (DOIs), facilitating subsequent retrieval of full-text content in publisher-specific formats. The list of DOIs of the research papers used and the names of their journals can be accessed from Zenodo[47].
(2) RedPajama. The RedPajama dataset[51], which served as the primary training corpus for the LLaMA2[29], encompasses diverse textual sources, including arXiv preprints, GitHub repositories, StackExchange discussions, Wikipedia articles and sanitized Common Crawl data. To preserve the model's foundational linguistic capabilities while preventing catastrophic forgetting, we extracted a representative subset of ~962 million tokens using LLaMA-2's sentencepiece tokenizer for pretraining of LLaMat-2 and ~805 million tokens using LLaMA-3's BPE tokenizer for pretraining of LLaMat-3. This strategic sampling maintains the model's general-purpose functionality while facilitating domain-specific knowledge acquisition.
(3) CIF. Despite the existence of multiple text-based crystal representations[19], CIF remain the definitive standard for structural data derived from diffraction studies. These standardized

files encode essential parameters, including unit cell dimensions, interaxial angles, space group symmetry operations and atomic position coordinates. Our dataset incorporates 470,000 CIF files, augmented with natural language descriptions generated via RoboCrystallographer[52]. These files were aggregated from three major sources: the Materials Project[53], GNoME-based ab-initio configurations[54] and the American mineralogist crystal structure database (AMCSD)[55].

(4) R2CID dataset integration. The integration protocol implemented a structured mixing strategy to optimize training efficiency and maintain model robustness. Research paper content was systematically interspersed with RedPajama text, maintaining a ratio of 1 million RedPajama tokens per 100 million research paper tokens. Crystallographic data integration occurred within the terminal 10% of the dataset, where CIF files and their descriptions were interleaved with research paper content.

Representativeness of pretraining dataset. Our pretraining corpus comprises peer-reviewed publications from approximately 500 Elsevier and 300 Springer journals, selected manually by domain-experts based on their relevance to materials science. While peer review provides a baseline quality filter, we acknowledge this does not guarantee complete absence of errors or biases in the literature. However, the aggregation of knowledge across ~4 million publications helps mitigate the impact of isolated incorrect claims through statistical averaging.

To ensure representativeness and prevent catastrophic forgetting of general capabilities, we implemented a carefully designed mixing strategy (see Supplementary Section E for details). Key aspects of this mixing strategy are mentioned below.

- Priority-based sampling: we manually categorized journals into three priority classes (P1, P2 and P3) based on materials science relevance. The highest priority class (P1) comprises 19 billion tokens, with P2 and P3 contributing 9 billion and 2 billion tokens, respectively (Supplementary Table E.1).
- Strategic interleaving: research papers were interleaved with RedPajama data at regular intervals (10 million research paper tokens followed by 0.1 million RedPajama tokens) to maintain general linguistic capabilities and reduce domain-specific overfitting.
- Specialized data integration: CIF (2.5% of corpus) and MSCD (0.019%) were incorporated in the final 10% of training to introduce specialized knowledge without overwhelming the primary corpus.
- Curriculum-based exposure: higher priority materials science content appeared with increased frequency towards later training stages, allowing the model to first establish broad scientific understanding before specializing.

Instruction fine-tuning datasets. The instruction fine-tuning protocol incorporated multiple specialized datasets encompassing materials science and general question-answering tasks. We developed two novel domain-specific datasets: MatBookQA, consisting of materials science questions and answers generated via GPT-4 using contextual prompting and a comprehensive question bank derived from the Graduate Aptitude Test in Engineering. The Graduate Aptitude Test in Engineering is a standardized examination for postgraduate admissions at premier Indian institutions and select international universities. The constituent datasets are detailed below.

(1) OpenOrca. The OpenOrca corpus encompasses 800,000 high-fidelity instruction-response pairs spanning diverse technical domains. Previous investigations[31] have demonstrated that models fine-tuned on this dataset exhibit superior performance across multiple evaluation frameworks, including Big-Bench Hard and AGIEval. This enhanced performance manifests in improved technical comprehension, complex query resolution and domain-appropriate response generation. Dataset optimization procedures were implemented to determine the optimal training sample size for our specific application (Supplementary Section E).

(2) Mathematics Corpus (MathQA). To enhance the model's quantitative reasoning capabilities, we incorporated 7,500 selected problems from the Math dataset[32]. This curated subset consists of advanced competition-level mathematical problems chosen to develop robust problem-solving abilities across various mathematical domains.

(3) Materials Science Instruction Sets (MatSciInstruct). The materials science instruction corpus integrates multiple specialised datasets, including a novel collection generated through GPT-4 (gpt-4-0613) using open-source materials science textbooks as source material. This approach generated contextually rich questions spanning diverse materials science subdomains. The corpus incorporates MatSciInstruct[34], which employs a two-phase development framework: an initial generation phase utilizing an instructor model to create domain-specific instruction data, followed by a Verification phase wherein a distinct verifier model assesses instruction quality across multiple dimensions including accuracy, relevance, completeness and logical consistency. The instruction set is further augmented with the MatSciNLP training corpus and our custom-developed MatBookQA dataset.

(4) MatBookQA. The MatBookQA dataset was systematically developed using a comprehensive materials science textbook[56]. The development protocol employed chapter-wise GPT-4 prompting using 20 distinct prompt templates (see Supplementary Section H for details), equally divided between generating short and extended responses. This methodology yielded 2,069 question–answer pairs, comprising 1,887 concise responses and 182 comprehensive explanations.

(5) Materials Science Question Answering (MaScQA). The MaScQA dataset encompasses 1,585 questions from Indian undergraduate engineering examinations, specifically 1,036 from civil engineering and 549 from chemical engineering curricula. Answer validation was performed using the GPT-4o model (2024-02-01), with only verified correct responses retained in the final dataset. As detailed in ref. 15, the question taxonomy includes four distinct categories: traditional multiple-choice, correlation-based matching, numerical multiple-choice and open-ended numerical problems.

(6) CIF dataset. To train the language models to generate crystals, we created a new set of tasks that enable the language models to train on various aspects of CIF. Specifically, we developed instruction–output pairs from CIF files sourced from AMCSD, Google GNoME and the Materials Project to enhance LLaMat's crystallographic comprehension and natural language query resolution capabilities. To this extent, we developed an instruction set implementing a dual-task framework comprising syntactic and semantic components. Syntactic tasks address the structural interpretation of CIF files. By contrast, semantic tasks, inspired by ref. 10, focus on crystal stability principles, including elemental co-occurrence patterns, atomic spatial distributions and stability-determining properties. This methodology generated approximately 7 million instruction–output pairs (6,941,865 training instances and 27,183 validation instances). The complete task framework, with corresponding system prompts detailed in Supplementary Section I, encompasses:

Syntactic analysis tasks:

- Atomic frequency quantification within crystal structures
- Spatial coordinate-based atomic identification
- Crystal parameter determination: dimensional analysis, volumetric calculation and space group classification
- Site occupancy equivalence evaluation
- Structure-based chemical formula derivation

Semantic analysis tasks:

- Property-conditioned crystal structure generation
- Positional atomic prediction using the `MASK` token methodology
- Structural dimension prediction for stability optimization
- Element-constrained crystal structure synthesis

MatNLP. The model evaluation employed a comprehensive dual-stage assessment protocol encompassing both materials science and general language capabilities. The primary evaluation phase compared multiple model iterations to optimize architectural decisions, whereas the secondary phase benchmarked performance against contemporary state-of-the-art materials science models. The primary evaluation corpus comprised 14 specialized materials science tasks, supplemented with four general-purpose reasoning and comprehension assessments to preserve broad linguistic capabilities.

Table A.3 and Supplementary Section B delineate the task taxonomy, dataset specifications and sample distribution across training and validation sets. The evaluation framework encompasses multiple task categories, namely, sentence classification, relation extraction, named entity extraction, synthesis action retrieval, paragraph classification, entity extraction, slot filling and question answering and multiple-choice question answering. Detailed task specifications are documented in Supplementary Section B and ref. 33. Model evaluation incorporated single-epoch fine-tuning on the training corpus before validation assessment to ensure instruction comprehension.

The secondary evaluation phase utilized the MatSciNLP dataset[57–77], which reformulates these tasks as multiclass classification problems. This meta-dataset enables direct performance comparison with existing materials science language models. To maintain evaluation integrity, distinct model instances were trained for each evaluation phase due to potential dataset overlap. Performance assessment followed the methodology established in ref. 34, implementing single-epoch training on a condensed training set followed by evaluation on a comprehensive 170,000 sample validation corpus. Task-specific examples are provided in Supplementary Section K.

MatSIE. The extraction of structured information facilitates automated data processing and machine-readable format conversion. Given the domain expertise and structured data comprehension acquired through instruction fine-tuning, LLaMat models were hypothesized to demonstrate robust performance in structured extraction tasks. To further analyse this capability, we performed the evaluation of the models using instruction–output pairs derived from four specialized datasets: (1) doping, (2) general materials, (3) MOFs[17] and (4) DiSCoMaT[38].

The initial three datasets focus on entity recognition and relationship extraction within materials science texts. The DiSCoMaT dataset provides annotated tables extracted from materials science publications. For the entity-relationship datasets, we developed six system prompts serving as prefixes to query–response pairs, where responses conform to standardized JSON schemas as established in ref. 17 (Supplementary Section G). The DiSCoMaTdataset, originally developed for alternative applications, was transformed to generate JSON-structured annotations suitable for the language models (see Supplementary Section G for format specifications).

Model development methodology

Continued pretraining. Continued pretraining was employed on both LLaMat-2 and LLaMat-3 models with optimized hyperparameters (Supplementary Sections C and D). The pretraining corpus underwent hierarchical prioritization based on materials science relevance (P1 > P2 > P3). This corpus integrated MSCD data and incorporated RedPajama subset to mitigate catastrophic forgetting, supplemented with 470,000 CIF for structural comprehension. The integration methodology employed a dual-phase mixing strategy:

- Primary phase: 90% of P1 content integrated with P2 and P3 datasets through stochastic shuffling
- Secondary phase: remaining 10% of P1 content combined with the CIF dataset through stochastic shuffling

The resultant dataset underwent final integration with RedPajama using a token-ratio methodology: approximately 0.1 million RedPajama tokens per 10 million materials science tokens. The details of the pretraining dataset, along with the number of tokens, are mentioned in Supplementary Table E.1.

Instruction fine-tuning. The LLAMAT-Chat models, initialized with corresponding LLaMat model weights, underwent tri-phase instruction fine-tuning:

- Phase I: single-epoch fine-tuning on OpenOrca dataset to establish general instruction-following capabilities
- Phase II: three-epoch fine-tuning on mathematical questions, optimizing quantitative reasoning capabilities. The limited dataset size enabled the observation of continuous validation loss reduction
- Phase III: single-epoch fine-tuning on an integrated corpus comprising MatSciInstruct, MatSciNLP, MatBookQA and MaScQA, focusing on materials science-specific instruction comprehension

Implementation utilized the Megatron-LLM framework with learning rate initialization at 2×10^{-6}, scaling to 2×10^{-5} over initial 10% iterations, followed by cosine decay. This protocol was replicated for LLaMat-2 and LLaMat-3 chat model development.

Task fine-tuning.

(1) LLaMat-Chat. The final development phase incorporated combined training on the training set of MatNLP and MatSIE datasets. This phase employed a 10^{-5} learning rate with cosine decay over two epochs. The intention of this stage was to familiarize the LLaMat-Chat models with a wide range of tasks relevant to materials research, including scientific natural language processing, structured information extraction and tabular information extraction. All the training data from these datasets were mixed to form a single task dataset on which the LLaMat-Chat models were fine-tuned.

(2) LLaMat-CIF. Crystal structure generation capabilities were implemented through PEFT[10]. Optimal LLaMat checkpoints underwent instruction fine-tuning using the dataset detailed in the 'Instruction fine-tuning datasets' section, with model selection based on minimal validation loss (Supplementary Fig. D.2). Comprehensive fine-tuning specifications and hardware configurations are documented in Supplementary Sections D.2 and D.3.

Baselines

To compare the performance of LLaMat with existing general-purpose models, we considered LLaMA, Gemini-1.5 Flash-8B and Claude-3 Haiku. Note that these models were chosen as they were the closest comparable ones in the respective families with LLaMat models in terms of the number of parameters. To assess the effect of fine-tuning, LLaMA models were evaluated both with and without fine-tuning.

Evaluation metrics

Loss function. The loss function used to train the models for continued pretraining, instruction fine-tuning and task fine-tuning is the cross-entropy loss.

MatNLP and MatSIE. Our evaluation protocol addresses the challenge of objective assessment for materials science question-answering tasks. We use a normalized string matching approach implemented in our evaluation script[78]. The evaluation involves three main steps: (1) normalizing model outputs by removing punctuation, articles and standardizing formatting; (2) maintaining a curated list of valid answers for each dataset; and (3) matching normalized model answers to valid answers using edit distance similarity. This normalization process handles common variations in response formatting while preserving semantic content. The edit distance matching helps account for minor variations in phrasing or spelling while maintaining objectivity. We maintain lists of valid answers that include acceptable variations for each question type. The script then finds the most similar valid answer for each model response based on string similarity.

It should also be noted that different task types require different evaluation approaches. For structured tasks like named entity recognition, we assign binary scores for each entity extraction, which enables computation of micro-F1, macro-F1 and accuracy metrics. For classification tasks, we match model outputs to predefined label sets. For multiple-choice questions, the system identifies the selected choice and compares it to the correct answer. Evaluation for doping, MOFs and general materials tasks were done in the same way as mentioned in ref. 17, but the evaluations were only performed on the outputs that could be parsed as json. No manual human evaluation was done. Also note that the tasks in these datasets had less support compared with the MatNLP tasks, as can be seen in Supplementary Section A.2. Evaluations on DiSCoMaT dataset was also evaluated similarly, with only outputs parsed as JSON format were considered. We calculated accuracy for each task separately, and the exact match criteria counts how many outputs matched exactly with the gold answers.

We acknowledge several limitations of this automatic evaluation approach. It may penalize semantically correct but lexically different answers, as the system relies on string matching rather than semantic understanding. The method cannot assess the quality of explanations or reasoning processes that models provide alongside their answers. In addition, responses that are contextually appropriate but phrased differently from expected answers may receive lower scores. Despite these limitations, this method provides consistent and reproducible evaluation across different models and is widely adopted in natural language processing evaluation. The approach enables systematic comparison of model performance while maintaining objectivity in assessment.

To evaluate the performance of models on the downstream tasks in MatNLP and MatSIE, precision, recall and F1 scores are used with the annotated data as the ground truth.

Crystal generation. For unconditional generation, we evaluate LLaMat-CIF unconditional crystal generation capabilities using metrics adapted from the LeMat-GenBench framework[43]. Our evaluation protocol assesses structural validity, thermodynamic stability and generative performance on 10,000 generated structures per model.

- Validity: structures must satisfy hard constraints: CIF readability, minimum interatomic distances >0.5 Å, and density bounds (0.01–25 g cm^{-3}). We report the percentage passing all validity checks.
- Uniqueness and novelty: uniqueness measures the fraction of structurally distinct crystals within generated sets using Short-BAWL fingerprints[43]. Novelty quantifies structures absent from LeMat-Bulk reference database[43], employing composition-matched StructureMatcher comparisons.
- Energy-based metrics: we compute thermodynamic properties using an ensemble of three machine-learned interatomic potentials: MACE-MP[44], UMA[45] and Orb-v3[46]. For each structure, we calculate: (1) formation energy (E_f), (2) energy above convex hull (E_{hull}) by comparing with the density functional theory-based convex hull from LeMat-Bulk, and (3) relaxation r.m.s. deviation between initial and relaxed structures. We report mean and standard deviation across the three machine-learned interatomic potentials. Lower values indicate greater thermodynamic favourability and structural stability.
- Stability: structures with $E_{hull} \leq 0$ eV per atom are classified as stable; those with $E_{hull} \leq 0.1$ eV per atom as metastable, accounting for synthesizable non-ground-state materials.
- SUN and MSUN rates. SUN counts structures that are stable ($E_{hull} \leq 0$), unique and novel, providing a composite benchmark. MSUN relaxes stability to include metastable structures ($E_{hull} \leq 0.1$ eV per atom). LLaMat-2-CIF required 13,000 generation attempts for 10,000 valid structures; LLaMat-3-CIF required approximately 33,000 attempts.

For conditional generation, we evaluated conditional generation on 9,046 test structures from ref. 10, where models generate crystals matching specified compositions and space groups.

- Extraction success measures the percentage of outputs successfully parsed as valid CIF files with extractable formulas.
- Match rate quantifies exact matches between generated and target compositions (element types and stoichiometric ratios).
- Space group matching assesses agreement with target symmetry specifications (international numbers 1–230). Performance is stratified by compositional complexity (element count, 1–6+) and crystal system (cubic, orthorhombic, tetragonal, monoclinic, hexagonal, trigonal and triclinic).
- The performance gap metric reports the absolute difference in Match Rate between LLaMat-2-CIF and LLaMat-3-CIF.

Additional metrics following ref. 10 methodology, including composition validity, recall and Wasserstein distance for element distributions (N_{el}) are reported in Supplementary Table J.1 for comparison with prior approaches.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The list of research papers used for creating the pretraining corpus and the crystal structures generated using the fine-tuned LLaMat models are available via Zenodo at https://zenodo.org/records/17251959 (ref. 47).

Code availability

The codes used in this work are available via GitHub at https://github.com/M3RG-IITD/llamat (ref. 42).

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Author contributions

N.M.A.K., M., M.Z. and S.M. conceived the project. V.M., S.S., M.Z. H.S.G. created the pretraining and fine-tuning datasets. V.M., S.S. and M.Z. performed the pretraining and fine-tuning. D.A. performed the final fine-tuning, inference and evaluations. B.M. supported the training on CS-2 Wafer Scale Engine. All authors contributed to writing and reviewing the work. S.M. was at Intel Labs when part of the work was done.

Competing interests

The authors declare no competing interests.

Additional information

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